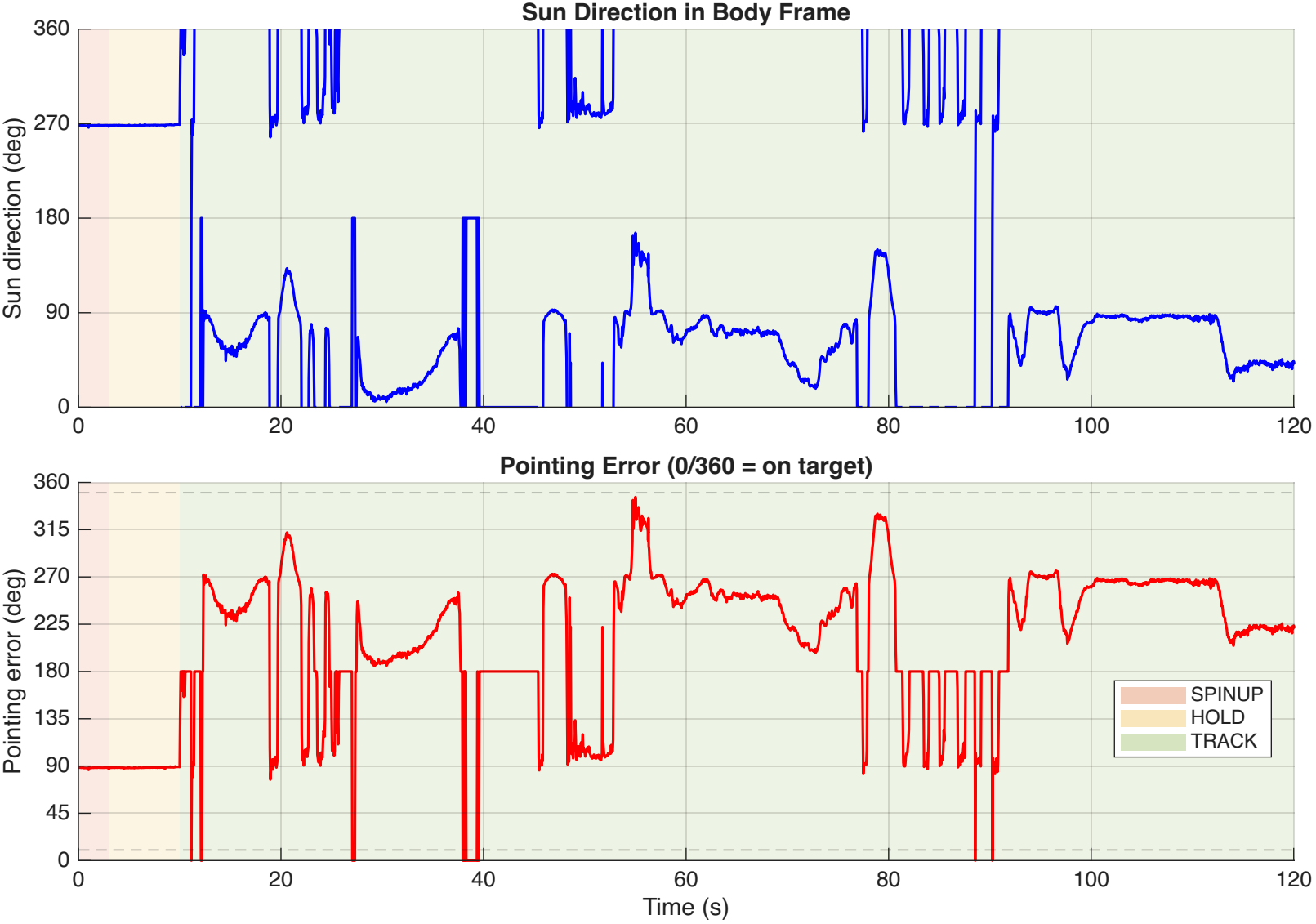
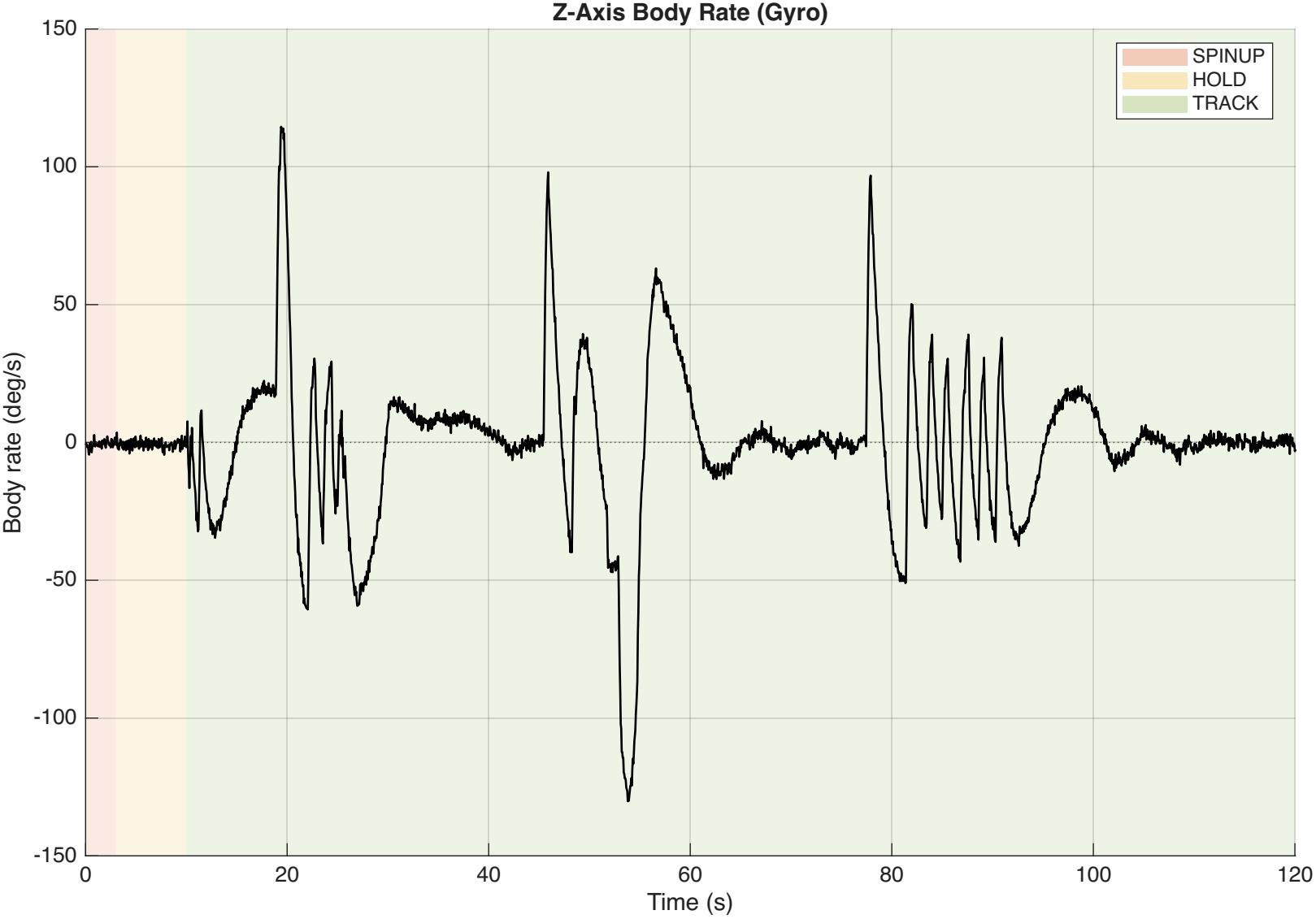
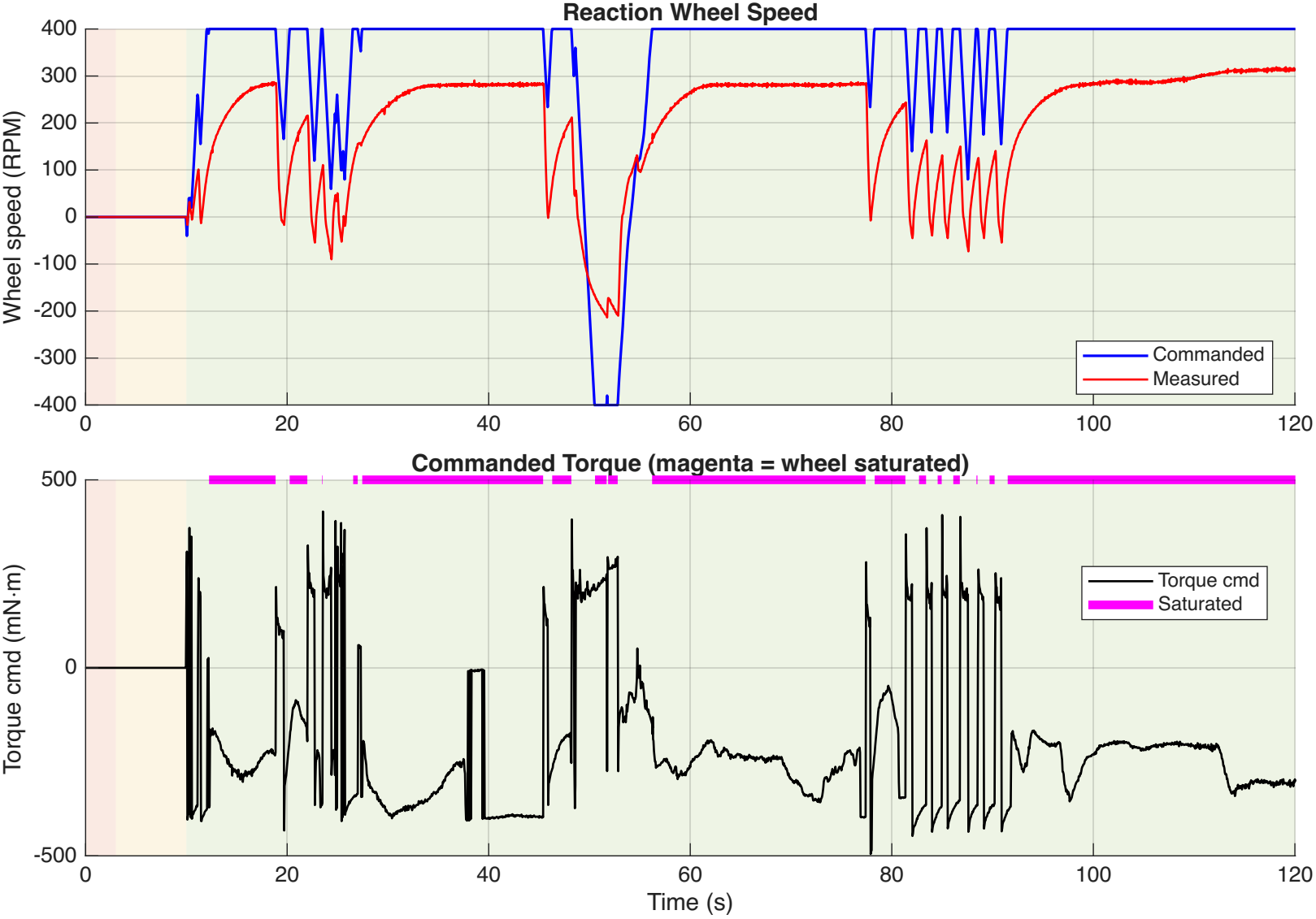
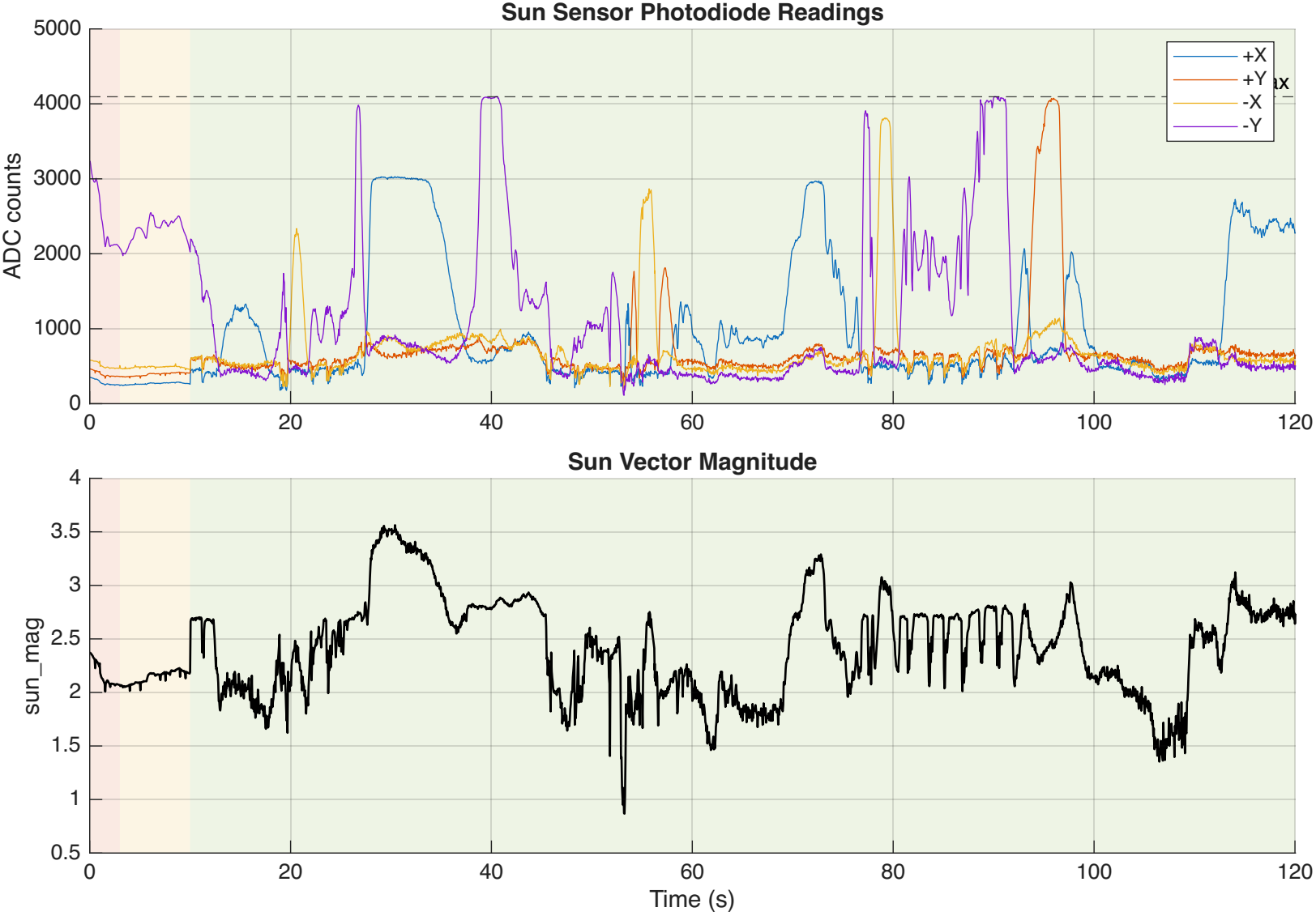
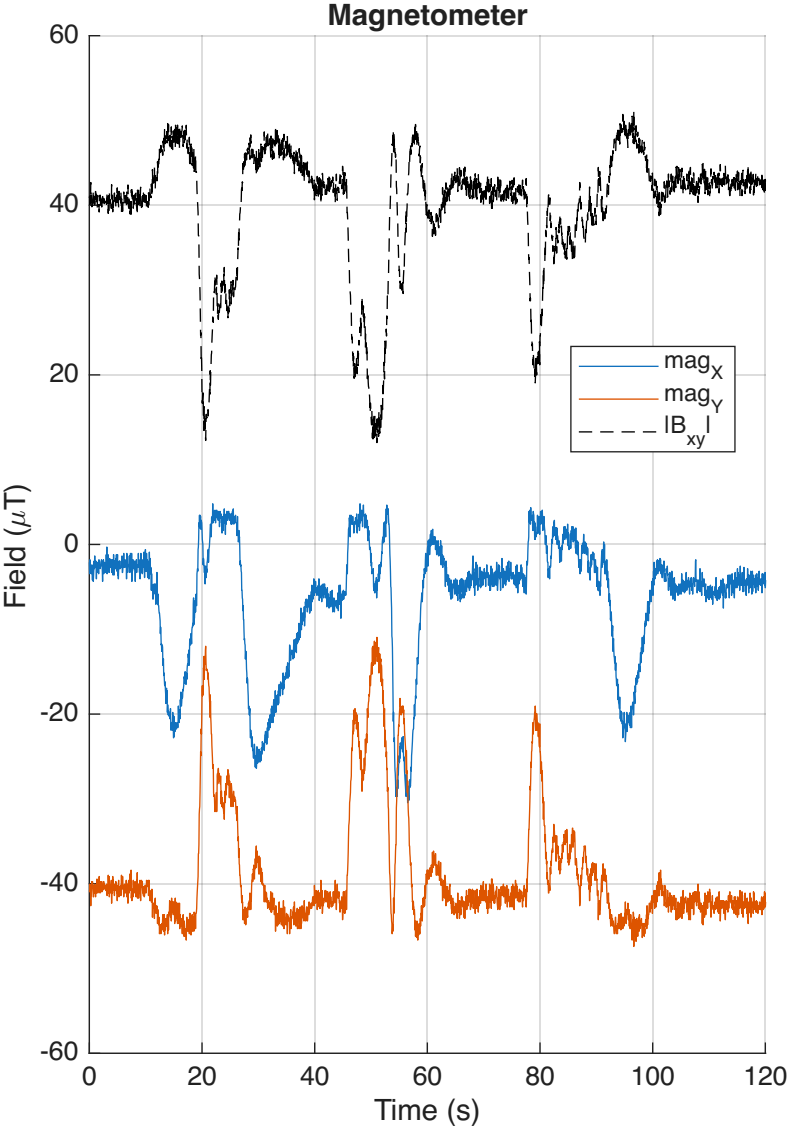
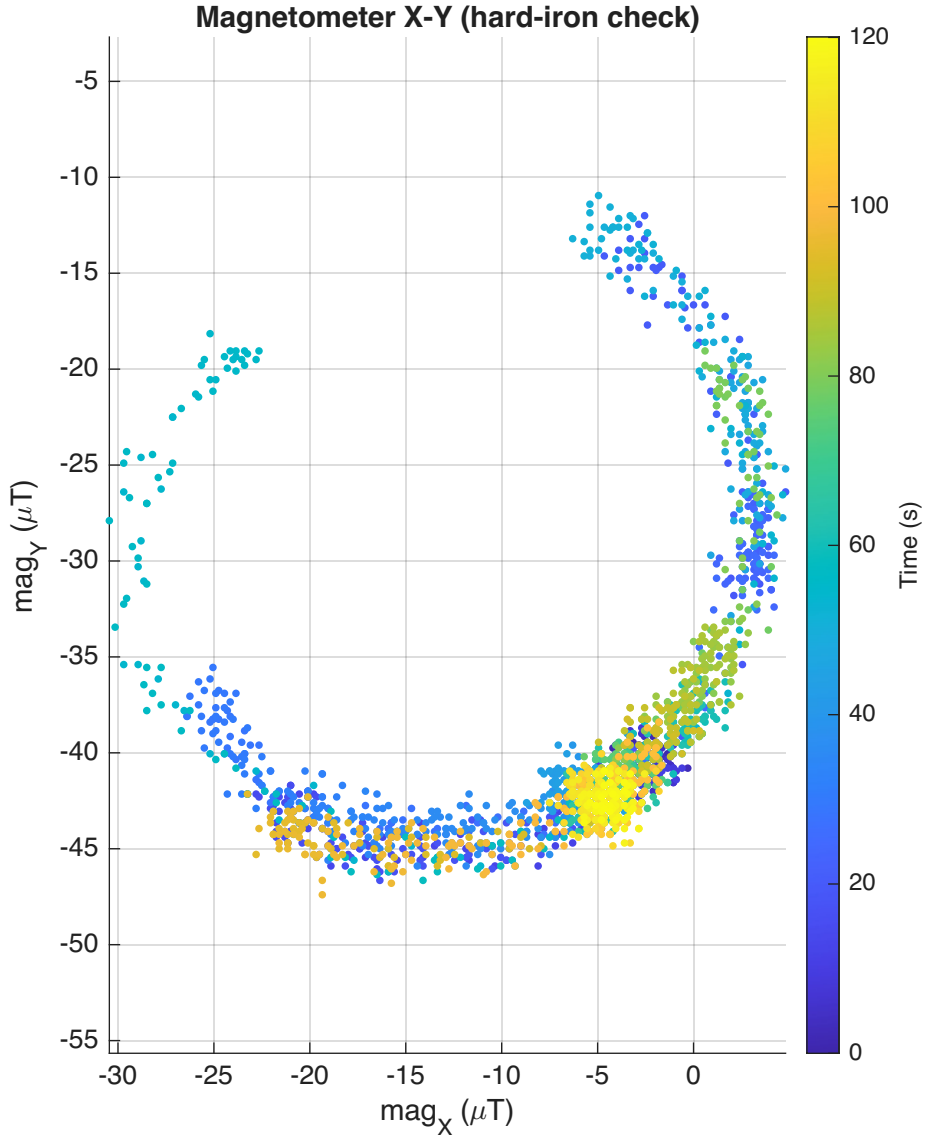

Sun Direction in Body Frame
Sun direction (deg)
360
270
180
90
0
0
20
40
60
80
100
120
Pointing Error (0/360 = on target)
Pointing error (deg)
360
315
270
225
180
135
90
45
0
SPINUP
HOLD
TRACK
Time (s)

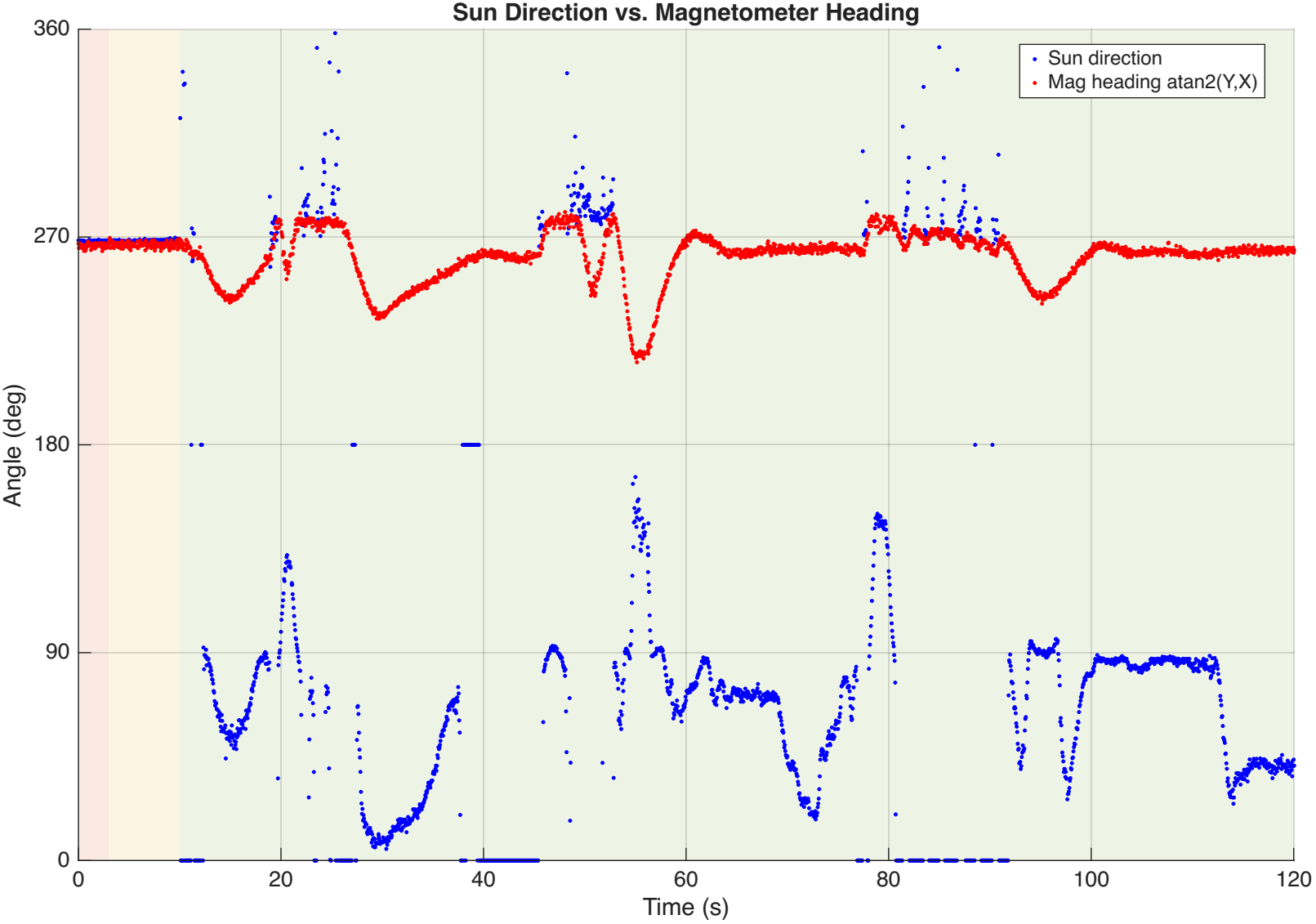

Z-Axis Body Rate (Gyro)
SPINUP
HOLD
TRACK
Body rate (deg/s)
150
100
50
0
-50
-100
-150
0
20
40
60
80
100
120
Time (s)

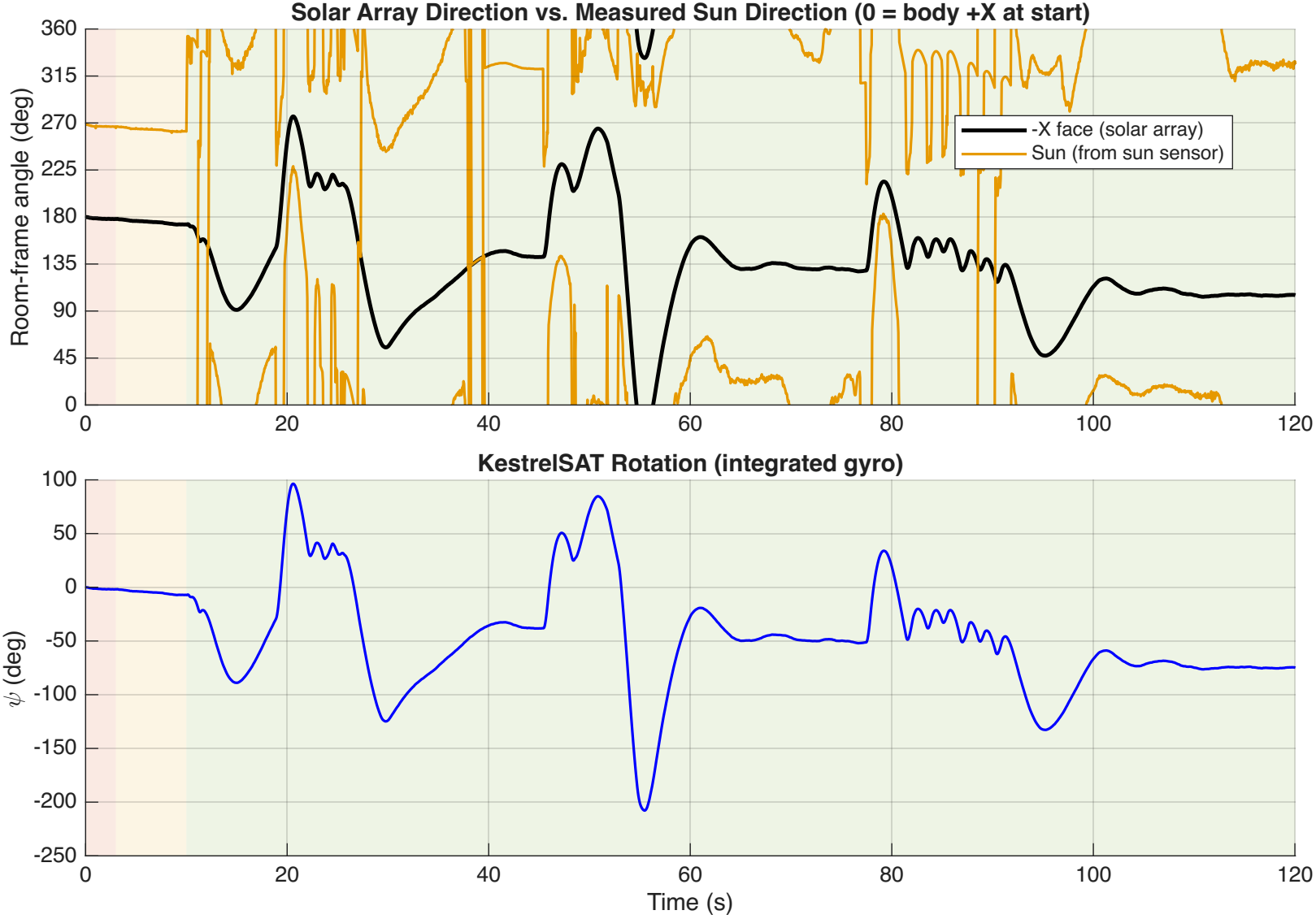

Reaction Wheel Speed
Wheel speed (RPM)
400
300
200
100
0
-100
-200
-300
-400
0
20
40
60
80
100
120
Commanded
Measured
Commanded Torque (magenta = wheel saturated)
Torque cmd (mN·m)
500
0
-500
Torque cmd
Saturated
Time (s)

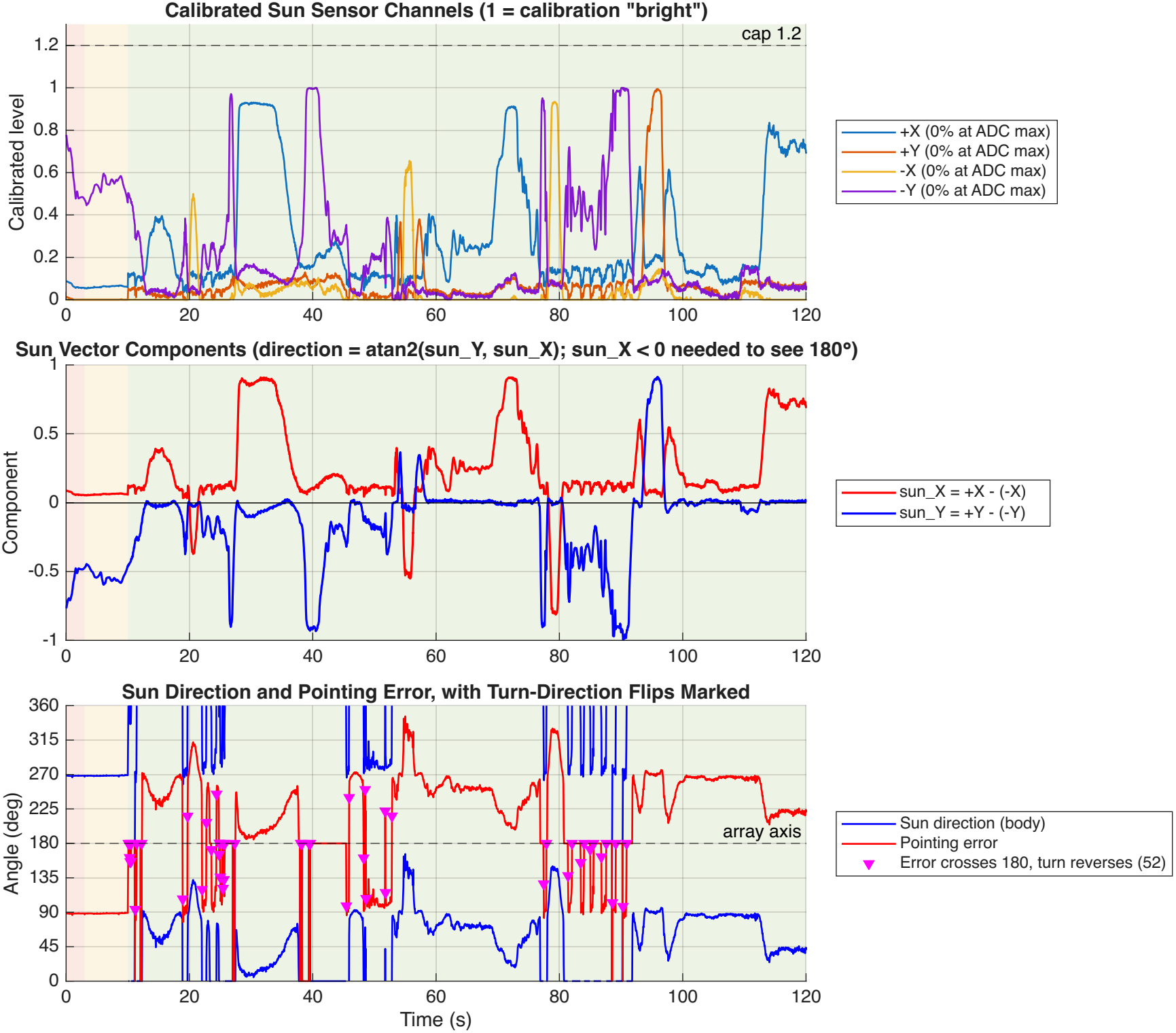

Sun Sensor Photodiode Readings
ADC counts
5000
4000
3000
2000
1000
0
0
20
40
60
80
100
120
+X
+Y
-X
-Y
Sun Vector Magnitude
sun_mag
4
3.5
3
2.5
2
1.5
1
0.5
Time (s)

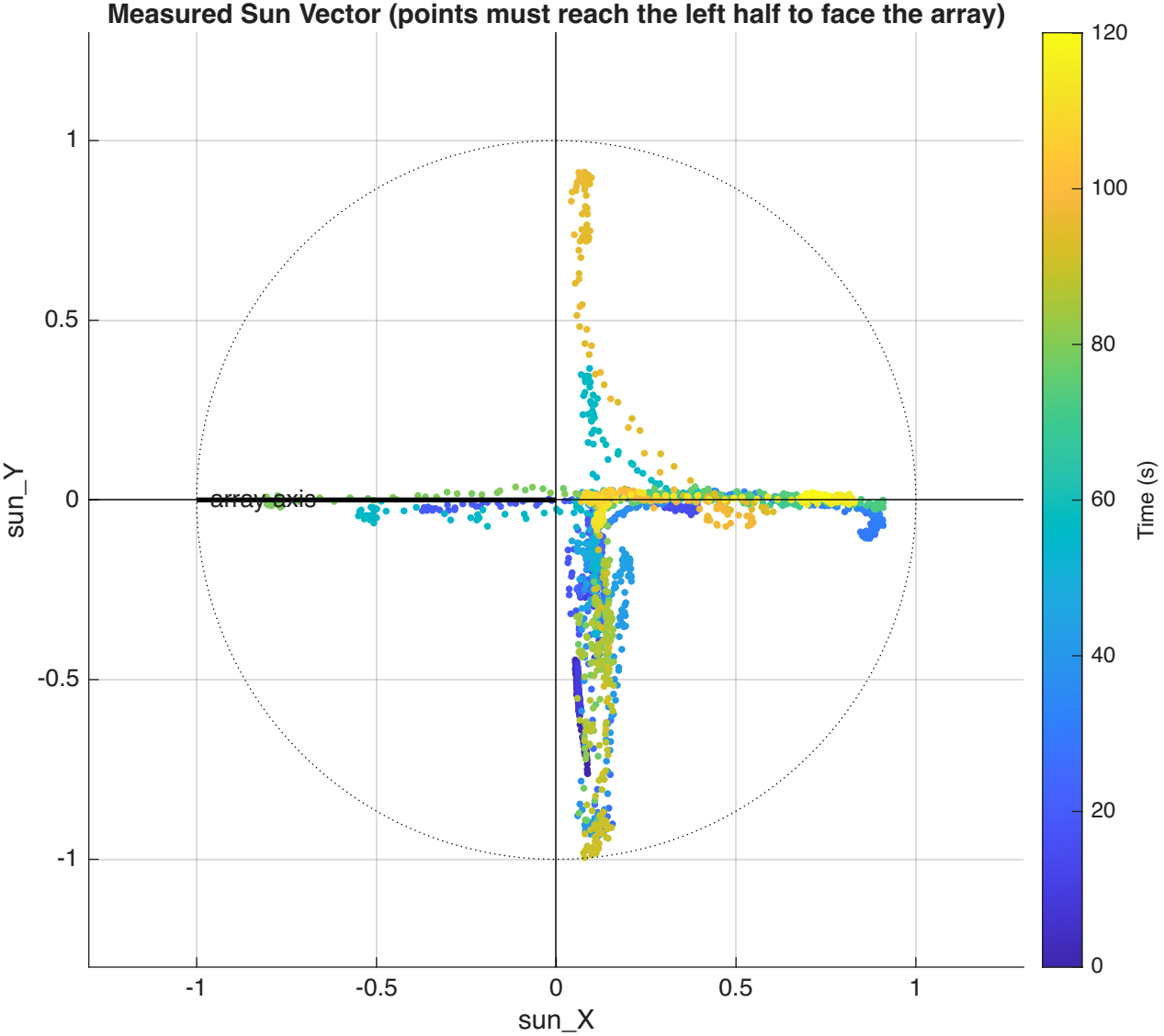

Magnetometer
Field (μT)
Time (s)
magX
magY
|Bxy|
60
40
20
0
-20
-40
-60
0
20
40
60
80
100
120


Magnetometer X-Y (hard-iron check)
magY (μT)
magX (μT)
Time (s)
-5
-10
-15
-20
-25
-30
-35
-40
-45
-50
-55
-30
-25
-20
-15
-10
-5
0
120
100
80
60
40
20
0

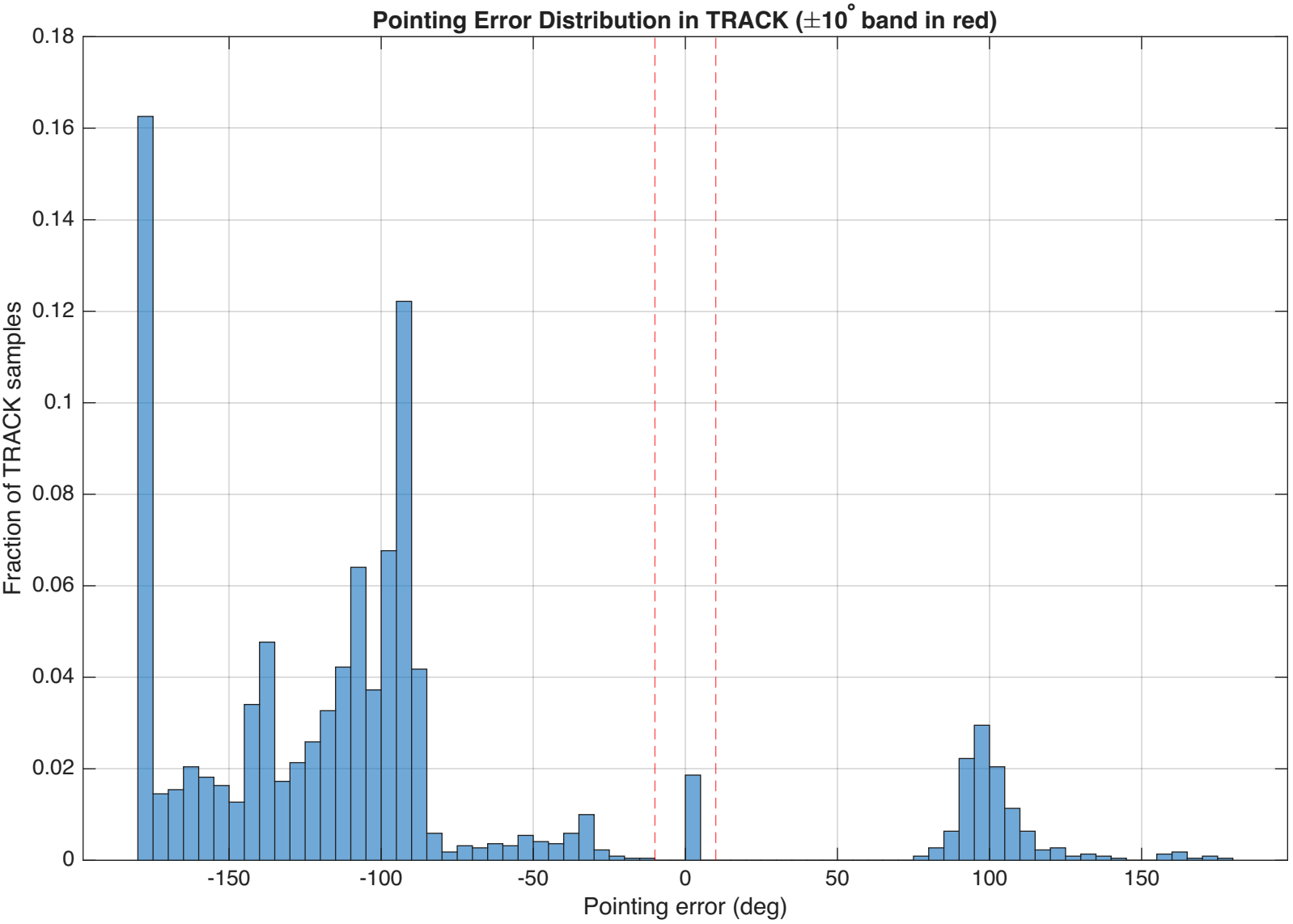

Sun Direction vs. Magnetometer Heading
Sun direction
Mag heading atan2(Y,X)
Angle (deg)
360
270
180
90
0
0
20
40
60
80
100
120
Time (s)


Solar Array Direction vs. Measured Sun Direction (0 = body +X at start)
Room-frame angle (deg)
360
315
270
225
180
135
90
45
0
-X face (solar array)
Sun (from sun sensor)
0
20
40
60
80
100
120
KestrelSAT Rotation (integrated gyro)
ψ (deg)
100
50
0
-50
-100
-150
-200
-250
Time (s)


Calibrated Sun Sensor Channels (1 = calibration "bright")
cap 1.2
Calibrated level
+X (0% at ADC max)
+Y (0% at ADC max)
-X (0% at ADC max)
-Y (0% at ADC max)
Sun Vector Components (direction = atan2(sun_Y, sun_X); sun_X < 0 needed to see 180°)
Component
sun_X = +X - (-X)
sun_Y = +Y - (-Y)
Sun Direction and Pointing Error, with Turn-Direction Flips Marked
Angle (deg)
array axis
Sun direction (body)
Pointing error
Error crosses 180, turn reverses (52)
Time (s)


Measured Sun Vector (points must reach the left half to face the array)
array axis
sun_Y
sun_X
Time (s)
1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
120
100
80
60
40
20
0


Phase Plane During TRACK (o = start, x = end)
Body rate (deg/s)
Pointing error (deg)
Time (s)
150
100
50
0
-50
-100
-150
-150
-100
-50
0
50
100
150
120
100
80
60
40
20


Pointing Error Distribution in TRACK (±10° band in red)
Fraction of TRACK samples
Pointing error (deg)
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
-150
-100
-50
0
50
100
150